Pranjali Jain

Phone: (820) 758-8234 | *Email:* pranjali_jain@ucsb.edu | *Pronouns:* she/her
github.com/PranjaliJain | linkedin.com/in/pranjali-jain | pranjalijain.github.io

Education

University of California, Santa Barbara (UCSB), CA — Sep 2023 – Present
Ph.D., Computer Science *(Expected Graduation: Aug 2027)* — GPA: **4.0/4.0**

University of California, Santa Barbara (UCSB), CA — Sep 2021 – Sep 2023
Master of Science (M.S.), Computer Science — GPA: **4.0/4.0**

Indian Institute of Technology (IIT), Gandhinagar, India — Jul 2016 – Aug 2020
Bachelor of Technology (B.Tech.) with Honors, Computer Science and Engineering — GPA: **8.55/10.0**

Awards and Honors

UCSB CS Outstanding PhD Student of the Year 2026, UCSB — Jun 2026
Distinguished Artifact Award, IISWC 2025 — Oct 2025
Best Paper Nomination, IISWC 2025 — Oct 2025
National Science Foundation (NSF) Research Grant for Sustainable Computer System Design — Jun 2025
IBM PhD Fellowship 2024-2025 Awardee — Jan 2025
Qualcomm Innovation Fellowship 2025 Finalist — Jan 2025
Socio-Environmental Knowledge (SEEK) Commons Fellowship 2025 Finalist — Jan 2025
Received **Travel Grant** to attend **ASPLOS 2024** — Mar 2024
Received **Travel Grants** to attend **CoDS-COMAD 2020** and **ECIR 2020** — Sep 2019, Feb 2020
Dean's List for Academic Excellence, IIT Gandhinagar — Jan 2017 – Dec 2019
Durga Devi Sultania Scholarship, IIT Gandhinagar — Apr 2019
Best Paper Award (one of three selected from 300+ submissions), ICSTEM 2019 — Jan 2019
Gujarat Government Award for **Social Good through CS Research**, ICSTEM 2019 — Jan 2019
Explorer's Fellowship, IIT Gandhinagar — Apr 2017
Conferred **Bachelor of Arts** degree in **Indian Classical Music** — Aug 2014

Experience

Graduate Research Assistant, *UCSB Archlab*, CA — Jun 2023 – Present
Mentor: Prof. Timothy Sherwood

- Collaborated with interdisciplinary teams to execute research on computer architecture and sustainable computing.
- Built computer architectural simulation frameworks utilizing state-of-the-art deep learning techniques.
- Evaluated emerging technologies as system hardware alternatives to improve performance and sustainability.

Research Intern, *Microsoft*, Redmond, WA — Jun 2026 – Sep 2026
Mentors: Rob Chappell, Rodrigo Fonseca

- Developed a framework to characterize Azure's fleet across microarchitectures using CPU performance telemetry.
- Evaluated representativeness of thousands of workloads across benchmark suites (SPEC CPU 2017/2026, DCPerf, DeathStarBench) against real fleet behavior, using coverage and proximity metrics.
- Contributed to constructing synthetic benchmarks to represent fleet behaviors not covered by existing suites.

Research Intern, *Microsoft*, Redmond, WA — Jun 2025 – Sep 2025
Mentors: Rob Chappell, Daniel Berger

- Designed Intel Core performance monitoring models for all microarchitecture generations present in the Azure fleet.
- Deployed models with Azure Telemetry Framework and collected performance monitoring data from 3.2M+ nodes.
- Developed data analysis methods and clustering techniques to characterize logical processor usage across nodes.
- Created a pipeline for workload characterization using performance monitoring telemetry from the Azure fleet.

Central Mentor, *UCSB Early Research Scholars Program*, CA Aug 2024 – Jun 2025

Mentor: Prof. Ziad Matni

- Managed a diverse group of 39 students across 10 teams, facilitating their research development over 3 quarters.
- Coordinated 10 year-long technical projects in collaboration with 18 faculty and graduate student mentors.
- Supervised the Research Methods course and supported development of research proposals and poster presentations.

Software Engineering Intern, *Qualcomm*, Santa Clara, CA Jun 2022 – Sep 2022

Mentor: Neeraj Sharma

- Developed an automated reliability testing framework for Qualcomm Wi-Fi chipsets using C# and internal tools.
- Designed testing algorithms for 10+ standalone and reliability scenarios and built a physical testbed to validate chipset performance across diverse operating conditions.

Embedded Software Engineer, *Enphase Energy*, Bangalore, India Aug 2020 – Feb 2021

Mentor: Sesha Narayanan

- Developed C++ features for the Envoy micro-controller platform and implemented APIs for real-time solar energy and system performance monitoring.
- Analyzed and remotely solved issues in the online platform or onsite installations of Enphase devices at 100+ sites.

Undergraduate Research Assistant, *IIT Gandhinagar*, India Aug 2018 – Aug 2020

Mentor: Prof. Joycee Mekie

- Designed and simulated 10+ server system cache designs with SRAM cache cells of different cell latencies.
- Devised 8 different last-level cache designs for mobile devices with conventional, NVM, and hybrid technologies.
- Evaluated performance, density, and energy benefits of different cache designs for 25+ workloads.

Research Intern, *SAFARI Group, ETH Zurich*, Switzerland May 2019 – Aug 2019

Mentor: Prof. Onur Mutlu

- Collaborated with a team of four to design C/C++-based algorithms for Efficient Computation of GPU Primitives (5+ operations) leveraging Lightweight Compute Units in L2 cache.
- Characterized 30+ primitive General-Purpose GPU applications to identify suitability for Processing-in-memory.

Summer Research Intern, *IIT Gandhinagar*, India May 2018 – Jul 2018

Mentor: Prof. Manu Awasthi

- Analyzed the impacts of changing 5+ timing parameters (e.g., read/write time) used to model DRAM-based volatile and STTRAM-based non-volatile memory systems using a C/C++-based main memory simulator ZSim-NVMain.
- Developed comprehensive documentation and a portable *Docker version* of the simulator.

PUBLICATIONS

Published paper titles are hyperlinked

Supporting Computer Architecture Ecological Impact Analysis. — *under review*
Pranjali Jain, Zachary D. Sisco, Dingsheng Li, Jonathan Balkind, Timothy Sherwood.

Environmental Trade-offs in AI Datacenter Cooling. — *under review*
Pranjali Jain, Mariam Elgamal, Gu-Yeon Wei, David Brooks, Gage Hills, Timothy Sherwood.

Characterizing the e-waste footprint of GPUs. — *under review*
Pranjali Jain, Stephen Browne, Josephine Wong, Mira Smith, Dingsheng Li, Jonathan Balkind, Timothy Sherwood.

The Environmental Footprint of AI Computing Systems. — *under review*
Demetrios E. Vandarakis, Pranjali Jain, Dingsheng Li, Timothy Sherwood.

Architecture for Energy Efficient Training using ZO Optimization. — *under review*
Naomi Rehman, Rhys Gretsch, Pranjali Jain, Eliah Reeves, Yequan Zhao, Jonathan Balkind, Zheng Zhang, Timothy Sherwood.

Matchmaker: Protocol Compatibility Checking for Heterogeneous SoCs. — *under review*
Yeganeh Aghamohammadi, Wenxuan Xu, Pranjali Jain, Vishal Seenivasan, Yuchen He, Benjamin Asperheim, Zachary D. Sisco, Jonathan Balkind.

Modeling PFAS Use in Integrated Circuits Fabrication. — *under review*
Mariam Elgamal, Pranjali Jain, Abdulrahman Mahmoud, Timothy Sherwood, Gu-Yeon Wei, David Brooks, Gage Hills.

Quantifying the Computing Energy Efficiency Paradox.
Pranjali Jain, Jonathan Balkind, Timothy Sherwood.
***ACM SIGEnergy Energy Informatics Review**, 2026.*
*Presented at the **Workshop on Sustainable Computer Systems (HotCarbon)**, 2026.*

Tracing the Computer System Supply Chain.
Pranjali Jain, Pranav Gunhal, Jonathan Balkind, Timothy Sherwood.
***Workshop on Ethical Systems and Architecture Design** at Architectural Support for Programming Languages and Operating Systems **(ASPLOS)**, 2026.*

Metallic toxicity as a consideration for future computing paradigms.
Pranjali Jain, Alexander Bologna, Dingsheng Li, Timothy Sherwood.
***npj Unconventional Computing**, 2026.*
*Presented at the **International Conference on Rebooting Computing (ICRC)**, 2025.*

Learning Architectural Cache Simulator Behaviour.
Pranjali Jain, Meiru Han, Zhizhou Zhang, Brandon Lee, Jonathan Balkind.
*IEEE International Symposium on Workload Characterization **(IISWC)**, 2025.*
Best Paper Nominee. Distinguished Artifact Award.

Leveraging Deep Learning for Architectural Cache Simulation.
Pranjali Jain, Meiru Han, Zhizhou Zhang, Brandon Lee, Jonathan Balkind.
*Workshop on Modeling and Simulation of Systems and Applications **(ModSim)**, 2025.*

Empowering E-Waste Recycling with Intelligent PCB Component Detection.
Pranjali Jain, Madison Long, Claire Pemberton*, Ivan Hernandez*, Mariana Rosillo*, Samantha West*, Jonathan Balkind, Timothy Sherwood.
*ACM SIGCAS/SIGCHI Conference on Computing and Sustainable Societies **(COMPASS)**, 2025.*

Electronic Waste footprint of Computer Systems.
Pranjali Jain, Claire Pemberton*, Ivan Hernandez*, Mariana Rosillo*, Samantha West*, Jonathan Balkind, Timothy Sherwood.
***Workshop on Hot Topics in Ethical Computer Systems** at Architectural Support for Programming Languages and Operating Systems **(ASPLOS)**, 2024.*

Toxicity versus Carbon Footprint of Computer Systems.
Pranjali Jain, Jonathan Balkind, Timothy Sherwood.
***Young Architect Workshop** at Architectural Support for Programming Languages and Operating Systems **(ASPLOS)**, 2024.*

HyGain: High Performance, Energy-Efficient Hybrid Gain Cell based Cache Hierarchy.
Sarabjeet Singh, Neelam Surana, Pranjali Jain, Joycee Mekie, Manu Awasthi.
*ACM Transactions on Architecture and Code Optimization **(TACO)**, October 2021.*

PlantDoc: Dataset for Visual Plant Disease Detection.
Davinder Singh*, Naman Jain*, Pranjali Jain*, Pratik Kayal*, Sudhakar Kumawat, Nipun Batra.
*ACM India Joint International Conference on Data Science & Management of Data **(CoDS-COMAD)**, 2020.*

NLPExplorer: Exploring the Universe of NLP Papers.
Monarch Parmar*, Naman Jain*, Pranjali Jain*, Jayakrishna Sahit*, Soham Pachpande*, Shruti Singh, Mayank Singh.
*European Conference on Information Retrieval **(ECIR)**, 2020.*

AgriBot: Agriculture-Specific Question Answer System.
Naman Jain*, Pranjali Jain*, Pratik Kayal*, Jayakrishna Sahit*, Soham Pachpande*, Jayesh Choudhary, Mayank Singh.
*14th International Conference on Science, Technology, Engineering and Management **(ICSTEM)**, 2019.*

** indicates equal contribution*

Research Projects

Greener Power Delivery for AI Datacenter Hardware | *C/C++, SystemVerilog* Apr 2026 – Present
Advisor: Prof. Timothy Sherwood | Collaboration with *UC San Diego* and *Google*

- Investigated removal of bulk capacitors from power delivery networks in datacenter servers and SSDs to reduce e-waste toxicity and board space overhead in AI accelerator infrastructure.
- Designed architectural mechanisms to tolerate voltage transients and power supply noise due to capacitor removal.

Sustainable End-of-Life Pathways for Computing Hardware | *Python* Mar 2026 – Present

Advisor: Prof. Timothy Sherwood

- Evaluated end-of-life pathways for computing devices spanning storage, refurbishment, recycling, and informal disposal, to understand their carbon, toxicity, and material recovery outcomes.
- Analyzed personal computing devices and AI data center fleets to inform hardware retirement decisions.

AI-based Malware Detection using Heatmaps | *C/C++, Python* Sep 2024 – Present

Advisor: Prof. Jonathan Balkind

- Designed and simulated custom hardware using *gem5* to generate modulo-memory access heatmaps on the fly.
- Developed AI-based dynamic malware detection system based on analyzing memory access patterns in real-time using modulo-memory access heatmap classification methods leveraging deep learning techniques.

PFAS Toxicity in Computer Manufacturing | *Python* Jun 2024 – Present

Advisor: Prof. Timothy Sherwood | Collaboration with *Harvard University*

- Conducted a comprehensive analysis to quantify the weight and toxicity impacts of per- and polyfluoroalkyl substances (PFAS) or Forever Chemicals of interest used in semiconductor manufacturing.
- Investigated trade-offs between PFAS toxicity and carbon emissions associated with computer system manufacturing.

Skills and Certificates

Languages – Python, C/C++, SQL, Shell scripting, C#, HTML/CSS, Ruby (Rails), Assembly Languages [MIPS]
Frameworks/Tools – Git, LaTeX, Python Data Science Stack, System Simulators [gem5, SniperSim], VS Code
Certificate in Inclusive Teaching, UCSB, Jun 2025
Graduate Student Mentor Training Course, Computing Research Association (CRA), Dec 2025

Teaching and Mentoring Experience

UR2PhD Program Teaching Assistant, *Computing Research Association (CRA)* Sep 2024 – Dec 2024

Early Research Scholars Program (ERSP) Mentor, *UCSB* Sep 2023 – Jul 2024

- Served as Technical Mentor for a team of 4 undergraduate students, guiding them in Computer Science research.
- Mentored the team to design an object detection system to identify and count components on printed circuit boards and supported the delivery of a final poster presentation and workshop paper submission.

Graduate Teaching Assistant, *UCSB* Sep 2021 – Sep 2024

- ***CS 8: Introduction to Computer Science*** Summer 2024
- ***CS 154: Computer Architecture*** Spring 2022 & Winter 2023 & Spring 2023
- ***CS 64: Computer Organization and Logic Design*** Winter 2022 & Fall 2022 & Spring 2024
- ***CS 170: Operating Systems*** Fall 2021
- ***RMP: Research Mentorship Program*** Summer 2024

Undergraduate Teaching Assistant, *IIT Gandhinagar* Jan 2019 – Aug 2020

- ***ES 215: Computer Organisation and Architecture*** Spring 2019 & Spring 2020

Professional Service

Invited Talk – Sustainability Impacts of Computer Systems, UCSB MRL Jun 2026
Invited Talk – Sustainability Impacts of Computer Systems, UCSD Apr 2026
Reviewer, COMPASS 2025 Late Breaking Work Apr 2025
Invited Talk – Toxicity Impacts of Computer Systems, U.Wash Sustainability Seminar Apr 2024
Reviewer, HPCA 2024 Artifact Evaluation Committee Nov 2023

Outreach

Graduate Representative, UCSB Women in Computer Science (WiCS) Aug 2023 – Sep 2025
High-School Outreach Initiative Coordinator, UCSB WiCS Sep 2023 – Sep 2024
Student Volunteer, Nyasa – Student-run social service initiative, IIT Gandhinagar Jul 2016 – Jun 2020